

Peer Response

by [Saleh Almarzooqi](#) - Monday, 1 September 2025, 7:13 AM

Hi Abdulrahman,

You hit the nail on the head about reasons why ACLs like KQML were invented: it is about going beyond sharing raw data (with all the uncertainty about what this data is) to sharing intentions and goals. I would agree with your argument that this higher-level communication is especially powerful in heterogeneous, open systems like e-commerce or supply chain networks. As Finin, Tim, et al. (1994) pointed out, the introduction of the performatives ask and achieve was an important step towards enabling agents to negotiate, cooperate, and collaborate across a range of platforms.

You also made an excellent comment regarding the practical limits of ACLs. The overhead of parsing semantics and the difficulty of sharing ontologies are non-trivial, particularly in time-sensitive environments. This accounts for the preference of many practitioners for simpler mechanisms such as APIs or method calls using Python or Java. As discussed by Kone, Shimazu, and Nakajima (2000), although ACLs are conceptually elegant, they are often too complex to be useful in a tightly managed environment.

From a different angle, it is reasonable to anticipate that ACLs will take on renewed importance as intelligent systems become increasingly distributed and autonomous (Soon, On, Anthony, and Hamdan, 2018). For example, modern multi-agent system frameworks such as JADE seek to simplify the story by including libraries that bring ACL concepts close to mainstream programming languages (Chaib-draa and Dignum, 2002). This hybrid approach may support a trade-off between the efficiency of method invocation and the semantic expressiveness of the ACLs.

ACLs look like a move in AI from the execution of tasks to real communication and negotiation. This angle lends itself to modern research in autonomous systems, where collaboration and adaptability are also important factors besides raw performance.

References

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Peer Response

by [Jaafar El Komati](#) - Wednesday, 3 September 2025, 8:42 AM

I liked how you explained that ACLs are about more than just passing data—they let agents communicate intent, which is crucial in open and distributed systems. Your example of supply chains and e-commerce really brought that to life, since those environments often involve independent agents that need to work together without being built on the same platform.

Still, as you pointed out, ACLs come with a lot of complexity. One way to prevent this could have been the use of lightweight semantic frameworks. By creating a smaller core set of performatives for everyday communication and keeping the more complex ones optional, systems could reduce processing overhead (Labrou, Finin and Peng, 1999). Ontology alignment tools could also help by resolving misunderstandings when agents interpret messages differently (Euzenat and Shvaiko, 2013). These tools allow agents to share meaning more easily without relying on manual fixes.

You also mentioned how developers often prefer simple method calls over using ACLs. A way to address that would be to better integrate ACLs into the programming environments developers are already comfortable with (Da Silva et al., 2019). If ACL libraries were as accessible and well-supported as typical API calls, it's likely they would have seen broader adoption.

All in all, your post does a great job of showing how ACLs move AI forward by allowing systems to understand and act on intent. With preventive steps like streamlined messaging, support for ontologies, and better developer tools, many of the challenges you mentioned might have been reduced.

References

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Peer response

by [Ahmed Ayman Abdelrahman](#) - Sunday, 7 September 2025, 7:46 AM

Thank you for your insightful discussion on Agent Communication Languages (ACLs). I agree with your point that ACLs go beyond simple data exchange by embedding meaning and intent, which makes them highly valuable in heterogeneous and open systems like e-commerce and supply chains. This “shared semantics” truly differentiates ACLs from traditional method invocation

I also found your observation about the trade-off between ACLs and conventional mechanisms such as APIs particularly important. While APIs provide speed and simplicity, ACLs support interoperability and autonomy—capabilities increasingly necessary as systems scale across organizational boundaries. As Jennings (2001) highlights, the ability of agents to negotiate, cooperate, and resolve conflicts depends on standardized communication protocols, something APIs alone cannot provide.

At the same time, your mention of performance limitations is crucial. In time-sensitive domains such as real-time trading or critical infrastructure monitoring, the overhead of parsing and interpreting ACL messages could indeed be a constraint (Russell and Norvig, 2020). This highlights the importance of context: ACLs shine in dynamic, decentralized environments, whereas simpler methods often remain sufficient in tightly controlled ones.

Overall, I think your post underscores how ACLs embody a shift from function execution toward true coordination and collaboration in AI systems. The challenge, as you point out, lies in balancing semantic richness with efficiency, depending on the system’s requirements.

References

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Russell, S. and Norvig, P. (2020) *Artificial intelligence: a modern approach*. 4th edn. Harlow: Pearson.